Temuulen Khishigsuren

Google Scholar | GitHub | Linkedin
Location: Melbourne, Australia

Email: kh.temulen@gmail.com
Phone: +61 430 064 832

RESEARCH INTERESTS

My research lies at the intersection of language, cognition, culture, and communication. I study how words and their meanings vary across languages and evolve over time, and the broader cultural, cognitive, and communicative factors that shape this variation and evolution. I use computational methods, including statistics and AI, together with information-theoretic approaches on naturalistic, large-scale data such as dictionaries and corpora to address these questions.

EDUCATION

University of Melbourne — Melbourne, Australia
PhD candidate in Psychological Sciences — *Aug 2023 – present*

National University of Mongolia — Ulaanbaatar, Mongolia
Master of Linguistics — *Sep 2020 – Dec 2022*

National University of Mongolia — Ulaanbaatar, Mongolia
Bachelor of Financial Management — *Sep 2009 – May 2013*

PUBLICATIONS

- **Temuulen Khishigsuren**, Francis Mollica, Ekaterina Vylomova, and Charles Kemp. (in press). Usage frequency predicts lexicalization across languages. *Nature Human Behaviour*.
- David Setiawan, **Temuulen Khishigsuren**, Pagnarith Pit, Aso Mahmudi, and Ekaterina Vylomova. (in press). MUDIDI: A two-stage framework for multilingual dictionary digitization with language models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.
- **Temuulen Khishigsuren**, Terry Regier, Ekaterina Vylomova, and Charles Kemp. (2025). A computational analysis of lexical elaboration across languages. *Proceedings of the National Academy of Sciences 122, no. 15*: e2417304122.
- **Temuulen Khishigsuren**, Francis Mollica, Ekaterina Vylomova, and Charles Kemp. (2025). Usage frequency predicts lexicalization across languages. In *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*, pp. 1458.
- Enkhjargal Purev, Oyunsuren Tsend, Purevsuren Bazarjav, **Temuulen Khishigsuren**. (2023). Color terms in Mongolian place names: A typological perspective. *Voprosy Onomastiki*. No. 1: 140-155.
- **Temuulen Khishigsuren**, Gábor Bella, Khuyagbaatar Batsuren, Abed Alhakim Freihat, Nandu Chandran Nair, Amarsanaa Ganbold, Hadi Khalilia, Yamini Chandrashekar, and Fausto Giunchiglia. (2022). Using linguistic typology to enrich multilingual lexicons: the case of lexical gaps in kinship. In *Proceedings of the 13th Language Resources and Evaluation Conference*, pp. 2798-2807.
- **Temuulen Khishigsuren**, Gábor Bella, Thomas Brochhagen, Daariimaa Marav, Fausto Giunchiglia, and Khuyagbaatar Batsuren. (2022). Metonymy as a universal cognitive phenomenon: evidence from multilingual lexicons. In *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*, pp. 2386-2393.
- **Temuulen Khishigsuren**, Gábor Bella, Thomas Brochhagen, Daariimaa Marav, Fausto Giunchiglia, and Khuyagbaatar Batsuren. (2022). How universal is metonymy? Results from a large-scale multilingual analysis. In *Proceedings of the 4th workshop on research in computational linguistic typology and multilingual NLP*, pp. 96-98.
- Gábor Bella, Khuyagbaatar Batsuren, **Temuulen Khishigsuren**, Fausto Giunchiglia. (2022). Linguistic diversity and bias in online dictionaries. *University of Bayreuth African Studies Online*, 173.

FELLOWSHIPS AND AWARDS

- PhD scholarship funded by Australian Research Council grant FT1910100200 (AU$140,000)
- General Institute Fellowship 2025, Linguistic Society of America (US$2,500); one of 12 recipients
- Postgraduate Research Excellence Award 2026, Melbourne School of Psychological Sciences (AU$1,000); first-place winner among the PhD cohort
- Undergraduate scholarship awarded by the Badraa Foundation, Mongolia, between 2010-2013
- One silver and two bronze medals in the second round of the Mongolian National Mathematics Olympiad

TEACHING AND SERVICE

- Teaching: one statistics and one research project subject at the Melbourne School of Psychological Sciences
- Supervision: one undergraduate and one master's student from University of Melbourne, Saarland University and Charles University
- Review for the Annual Meeting of the Cognitive Science Society
- Board member at Badraa Foundation, Mongolia, since 2013

CONFERENCE TALKS AND PRESENTATIONS

- Conference talk (2025) "Usage frequency predicts lexicalization across languages" at *the 47th Annual Meeting of the Cognitive Science Society*; selected for an oral talk among 14% of submissions
- Conference presentation (2022) "Metonymy as a universal cognitive phenomenon: evidence from multilingual lexicons" at *the 44th Annual Meeting of the Cognitive Science Society*.
- Conference presentation (2022) "Using linguistic typology to enrich multilingual lexicons: the case of lexical gaps in kinship" at *the 13th Language Resources and Evaluation Conference*.

MEDIA COVERAGE

- The Conversation article (2025): "Do Inuit languages really have many words for snow? The most interesting finds from our study of 616 languages" by Charles Kemp, Ekaterina Vylomova, Temuulen Khishigsuren, and Terry Regier; 34,790 reads.
- Scientific American article (2025): "Linguists find proof of sweeping language pattern once deemed a 'hoax'" by Cody Cottier and Allison Parshall
- Language Log article (2025): "Linguistic relativity: snow and horses" by Victor Mair
- Linguistic Discovery article (2025): "Do Inuit languages really have more words for snow? And why does it matter, anyway?" by Daniel W. Hieber

TECHNICAL SKILLS

Statistics	:	Frequentist and Bayesian statistics, Phylogenetic comparative methods
Languages & DB	:	R, Python, PostgreSQL
ML & AI	:	PyTorch, Hugging Face Transformers, Agentic Harness Optimization
Tools & Cloud	:	Git, Google Cloud Platform, Vertex AI

INDUSTRY EXPERIENCE

Consultant Mar 2022 – Jan 2023
Good Growth Co. *Ulaanbaatar, Mongolia*

- UK-based sustainability-driven company working with communities and partners to design and implement nature-positive value chains and place-based regeneration programs
- Role: produced analytical frameworks and documentation to evaluate and structure place-based regeneration programs and their integration into value chains; conducted financial modeling to assess the viability of value chains in the cashmere industry

Associate Apr 2012 – Jul 2018
Mongolia Opportunities Fund *Ulaanbaatar, Mongolia*

- The first-ever Mongolian private equity fund with IFC, EBRD, DEG, FMO, Mitsubishi Corporation, and ORIX as investors
- Role: performed industry analysis on various industries such as agriculture, banking and finance, health, and transportation; conducted detailed company research from a commercial perspective, including competitive landscape, SWOT, growth opportunity, risk assessment, financial analysis and modeling; wrote investment documents on several deals over US$10 million; implemented ongoing monitoring of one of the portfolio companies in banking and financial sector